CGD-460 Assignment 2: Stormwatch Harbour

Saad ul Irfan | Student ID: 231713 | CGD-7B | Experimental report, 30 October 2026

Submission status: The 30-seed rule-based study is complete. LLM and injection evaluations were interrupted due to local inference time. Incomplete experiments are identified rather than presented as finished.

A. Agent specification

Role	Performance / Environment	Actions / Sensors
Dockworker	Cargo delivered, safe movement; grid, ships, storms	Move, pick up, drop; nearby berth, assignment, remembered warn
Lookout	Timely warnings; approaching storms	Warn workers; storm forecast five ticks early
Quartermaster	Cargo throughput, fair berth use; ships and workers	Assign berth; berth status and worker availability

The harbour is partially observable (workers cannot directly sense storms), stochastic (ship arrivals vary), sequential (actions affect later outcomes), dynamic (ships and storms change), discrete (grid and ticks), and multi-agent (16 workers plus coordinating agents).

Chosen designs: reflex workers respond cheaply to immediate conditions but lack memory; model-based workers retain warnings and berth state at greater state-management cost; utility-based assignment scores cargo, distance, crowding and storm risk, offering richer decisions but requiring weight selection. Each role should be limited to its own duties: workers cannot authorize cargo release, the Lookout cannot close the harbour, and the Quartermaster cannot bypass safety approval. Greater autonomy increases coordination and safety risks.

B. Simulation and evaluation

The implemented grid is 24 by 16 cells, with four north-edge berths and a south-west warehouse. Student ID ending in 3 requires 16 workers, arrivals every 5-9 ticks, five storms, and nine cargo units per ship. Each run lasts 600 ticks; a seeded generator controls randomness.

Policy	Cargo mean ± SD	Ship wait mean	Crowding mean	Knockouts mean
baseline	0.00 ± 0.00	271.223	0	0
reflex	151.73 ± 99.03	204.887	0	0.4
model	162.00 ± 69.69	209.725	0	1.967
utility	201.37 ± 4.09	210.122	0	2.3

Table 1. Thirty-seed evaluation (seeds 11-40). Utility has the highest mean delivered cargo and much lower run-to-run cargo variation than reflex and model-based. No crowding violations were recorded; the occupancy counter still warrants a code-level audit.

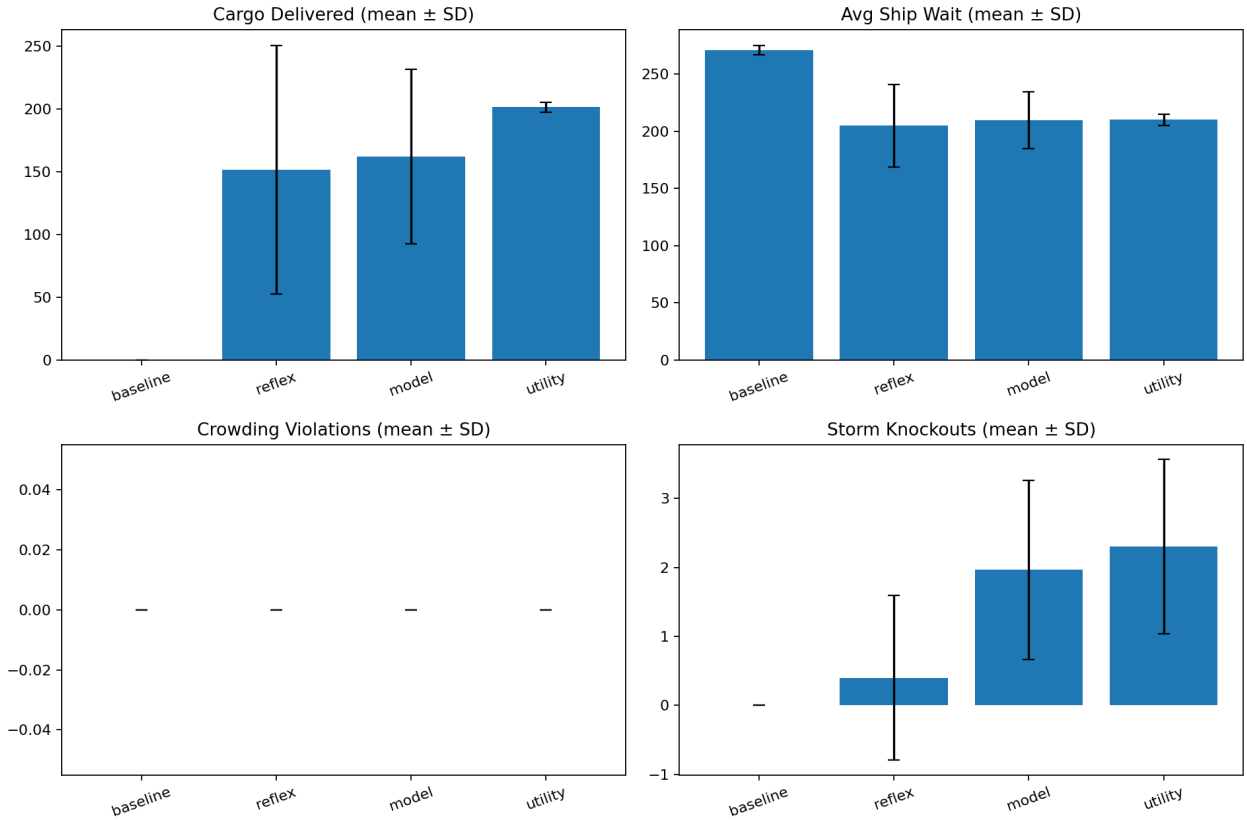


Figure 1. Policy comparison generated from the saved 30-seed results.

Emergence experiment: changing the movement tie-break reduced western-two-column visits from 61.59% to 27.68%, while mean delivered cargo fell from 200.9 to 190.9 over ten development seeds. This suggests a traffic concentration and throughput trade-off; the route rule, rather than explicit lane instructions, generated the observed pattern.

C. LLM Harbour Master

The local model was Qwen 2.5 3B via Ollama, temperature 0. The simulation sends a text-only harbour report every 20 ticks. The program parses proposed actions, validates them, executes permitted orders, logs results, retries malformed output at most twice, and uses a rule-based fallback. Prompts and responses are saved with ticks in logs/llm_log.jsonl.

Seed	Utility cargo	LLM cargo	LLM calls	Invalid outputs
11	196	188	38	8
12	202	205	40	12
13	193	191	38	9
14	201	185	33	3

Table 2. Completed paired LLM trials only (seeds 11-14). Seed 15 LLM run was interrupted. The LLM does not show a consistent cargo advantage in these completed pairs. Model calls are much slower than rule-based simulation.

The required ten-statement hallucination audit and three deliberate probes are not yet completed. Claims about hallucination rate cannot be made from the available evidence.

D. Prompt design and injection tests

The prompt design separates role, context, goals, constraints, and output format. The action schema is fixed, with validation against impossible or privileged actions. The initial injection experiment covers ship manifests, player messages, fabricated ship queries, and crew notes. Both instruction-boundary marking and executor permission checks are intended safeguards.

The saved attack log contains 10 completed trials of the planned 60 (10 cases × 3 trials × 2 conditions). These partial trials are not sufficient to claim a complete before/after attack success rate. The zero-shot/few-shot comparison and prompt change log are also outstanding.

E. Oversight and critical analysis

Assigning workers and issuing warnings are suitable for automated operation with monitoring because they are comparatively reversible. Holding a ship deserves monitoring and review when delays become costly. Closing the harbour and releasing cargo require human-in-the-loop approval because they have substantial operational impact. A smoke test confirmed rejection of an unapproved privileged action; the current CLI approval flag remains a testing bypass rather than a per-action interactive approval interface.

Based on the recorded performance and inference costs, a rule-based or hybrid approach is preferable to giving the LLM unrestricted control. The utility policy is consistent and fast; the LLM adds language-based planning but produces invalid outputs and substantial latency. Limitations include only four completed paired LLM seeds, incomplete attack trials, and evaluation of one local model at one temperature.

Reproducibility and references

Run: `python run.py --agent utility --seed 11`. The source repository includes run.py, src/, results/, logs/, requirements.txt, and README.md. Reported measurements come from the included CSV/JSONL files. References: CGD-460 Assignment 2 brief; Russell and Norvig (2021), Artificial Intelligence: A Modern Approach; Yao et al. (2022), ReAct.